

Jeep Cherokee Uconnect Hack (2015)

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Introduction

Fiat Chrysler Automobiles (FCA) deployed the Uconnect infotainment system to provide navigation, phone integration and entertainment. Because vehicles became internet-connected, new attack surfaces emerged: remote code execution, denial-of-service, and man-in-the-middle attacks. Unlike typical IT breaches, successful vehicle compromises can produce direct physical harm.



Background — The 2015 Demonstration

In July 2015 security researchers Charlie Miller and Chris Valasek exploited a zero-day in the vehicle's cellular-connected interface to reach the CAN bus. They demonstrated remote control of engine functions, braking and steering during a highway test near St Louis, Missouri. The incident prompted FCA to recall approximately 1.4 million vehicles.

Technical target: insufficient network isolation between infotainment and safety-critical ECUs on the CAN bus.



Social Issues — Public Safety's Trust

Erosion of Consumer Trust

Consumers may perceive connected features as liabilities, slowing adoption of advanced mobility technologies.

Privacy Infringement

Infotainment access can expose location history, contacts and microphone audio—risks include stalking and unwanted

Public Safety Risks

Remote manipulation of vehicle controls creates immediate risk to occupants and other road users — the hack could cause multi-vehicle collisions.



Social Issues — Financial & Emotional Costs

The recall affected ~1.4 million vehicles, generating direct costs (repairs, dealer visits) and indirect losses (reduced resale value, reputational damage). Owners faced time-consuming remediation and potential loss of confidence in manufacturer safety claims.

Emotional consequences included trauma for occupants involved in demonstrations or incidents, increased anxiety about using connected systems, and long-term distrust among prospective buyers.



Ethical Issues — Disclosure, Consents Responsibility

1

Responsible Disclosure vs Public Endangerment

The highway demonstration risked bystander safety; debate centres on whether public proof pressured faster fixes or unduly endangered uninvolved people.

2

Lack of Informed Consent

Only the journalist involved consented; Kantian ethics highlights that using uninformed individuals as means is unethical.

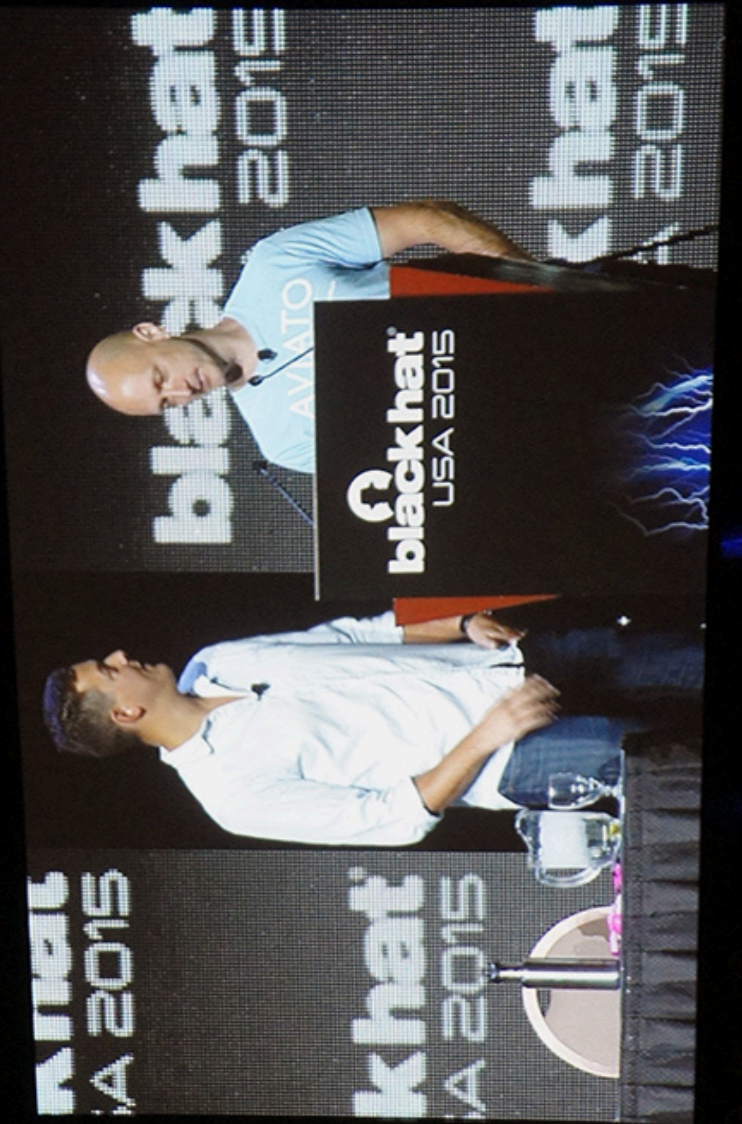
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Manufacturer Negligence

Failure to separate internet-facing systems from safety-critical controls reflects poor prudence and breach of professional duty.

Ethical Analysis — Perspectives Compared

Utilitarian view: public demonstration may accelerate fixes (greater good) but carries immediate risk. Deontological view: violating non-consenting parties' autonomy is impermissible. Virtue ethics: engineers and organisations should exhibit prudence and responsibility — prioritising safety over market features.



Legal s Professional Issues

- ♦ **Computer Fraud s Abuse** — unauthorised access could attract charges under statutes such as the CFAA; reckless conduct may lead to criminal liability.
- ♦ **Product Liability** — failure to design adequate isolation may constitute a defect, exposing FCA to civil suits for negligence and damages.
- ♦ **Regulatory Compliance** — potential breaches of safety standards (NHTSA) and consumer protection rules (FTC) risk fines and enforcement actions.
- ♦ **Professional Codes** — violations of ACM/IEEE principles (avoid harm; ensure safety) and BCSD duty of care raise questions of accountability and licencing standards.



Conclusion And Personal Reflection

Key takeaways: connected vehicles require rigorous security-by-design, explicit separation of safety and convenience systems, clear responsible-disclosure norms, and stronger regulatory oversight. Professional ethics demand prioritising public safety over commercial features.

Personal reflection: As a computing student, the case underscores the need to combine technical competence with ethical reasoning and stakeholder empathy when designing cyber-physical systems.

